Heterogeneous Graph Attention Network for Drug-Target Interaction Prediction

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ABSTRACT

Identification of drug-target interactions (DTIs) is crucial for drug discovery and drug repositioning. Existing graph neural network (GNN) based methods only aggregate information from directly connected nodes restricted in a drug-related or a target-related network, and are incapable of capturing long-range dependencies in the biological heterogeneous graph. In this paper, we propose the heterogeneous graph attention network (HGAN) to capture the complex structures and rich semantics in the biological heterogeneous graph for DTI prediction. HGAN enhances heterogeneous graph structure learning from both the intra-layer perspective and the inter-layer perspective. Concretely, we develop an enhanced graph attention diffusion layer (EGADL), which efficiently builds connections between node pairs that may not be directly connected, enabling information passing from important nodes multiple hops away. By stacking multiple EGADLs, we further enlarge the receptive field from the inter-layer perspective. HGAN advances 15 state-of-the-art methods on two heterogeneous biological datasets, achieving the results near to 1 in terms of AUC and AUPR. We also find that enlarging receptive fields from the inter-layer perspective (stacking layers) is more effective than that from the intra-layer perspective (attention diffusion) for HGAN to achieve promising DTI prediction performances. The code is available at https://github.com/Zora-LM/HGAN-DTI.

CCS CONCEPTS

• **Computing methodologies** → **Neural networks**; • **Applied computing** → **Bioinformatics**.

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KEYWORDS

Drug-target interaction prediction, graph neural network, heterogeneous graph

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1 INTRODUCTION

Identification of drug-target interactions (DTIs) is fundamental for drug discovery (i.e., to find novel drugs) and drug repurposing (i.e., to find new indications for existing drugs) [6, 38]. It becomes more urgent than ever before as a variety of diseases (e.g., Alzheimer disease and cancers) still cannot be cured and many new diseases (e.g., COVID-19) emerge every year [12, 19, 37]. Currently, only a small number of DTIs have been experimentally validated [5, 6], whereas, the number of all possible drugs (small molecules) has reached 111 million [22]. Traditional experimental verification of DTIs is time-consuming and expensive, and limited to a small scale [6]. Many studies have resorted to computational methods to identify drug–target associations effectively in a large scale. Benefiting from the rapid development of biological and chemical technologies, predicting potential DTIs from heterogeneous biological data using machine learning (ML) based methods has become the promising strategy [6, 30, 35, 38].

Recently, graph neural networks (GNNs) have achieved promising results in node/graph representation learning by utilizing both node attributes and the graph topology. They have drawn considerable research interest in a variety of fields [40, 53, 56]. A few GNN-based methods [20, 46] have been proposed for DTI prediction using heterogeneous biological data. They formulate heterogeneous biological data into multiple drug- and target-related networks (Figure 1(a)), where nodes are various bioentities (i.e., drugs, targets, diseases, side-effects, etc.) and edges are interrelations between these bioentities. Then they treat DTI prediction as a link prediction problem. However, these methods have shortcomings in the following

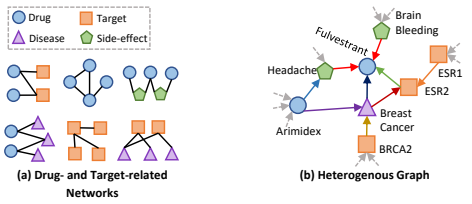

Figure 1: (a) Drug-/target-related networks. (b) Biological heterogeneous graph centered around drug Fulvestrant, where edges with different colors denote different relations, and arrows denote the direction of information transmission.

aspects. **Firstly**, they consider each relation network (e.g., *drug-disease* network) separately in an identical manner. Particularly, NeoDTI [46] applied graph convolutional networks (GCNs) [23] to each relation network, and IMCHGAN [27] applied graph attention networks (GATs) [45] to each relation network. Afterwards, bioentity node representations learned from different relations are summed up. However, they ignore the structural dependencies between different relation networks as well as the diverse semantic relations in cross-networks, e.g., *drug-target-disease*, which limits their capabilities in distinguishing the structural role of different bioentities. **Secondly**, existing works, for example, NeoDTI [46], are restricted to the receptive field of one-hop neighboring nodes and are incapable of capturing long-range dependencies in a single GNN layer. Although stacking multiple GNN layers increases the receptive field, such a deep model suffers from the bottleneck of feature smoothing [4]. Consequently, the DTI prediction performance is compromised. **Thirdly**, GCN and GAT are built for homogeneous graphs. Their operation is based on the assumption of homophily [13], i.e., connected nodes tend to have similar labels. Heterogeneous biological data in essence form a heterophily graph (biological heterogeneous graph) (Figure 1(b)) characterized with complex structures, distinct node types and edge types, which brings challenges for bioentity node representation learning. Therefore, directly applying GCN/GAT on biological relation networks inevitably restricts graph structure learning capability. Heterogeneous GNNs have presented encouraging performances in recommendation systems [3], job mobility prediction [53], and so forth. Nevertheless, extracting bioentity node representations for DTI prediction using heterogeneous GNNs are rarely explored.

In this paper, we propose a heterogeneous graph attention network (**HGAN**) to predict DTIs. To address the first and the third issues, we construct a biological heterogeneous graph with both node information and edge type information (Figure 1(b)). Such a graph description of heterogeneous biological data not only gives a systematic understanding of complex associations between bioentities, but also enables the integration of knowledge towards precise DTI prediction. Then, in order to eliminate the heterogeneous gap, we apply the node type-specific transformation to project bioentity node features into the same latent feature space. Similarly, we implement the edge type-specific transformation to project edge type features into the same latent feature space.

In order to tackle the second issue, we enlarge the receptive fields of HGAN on the biological heterogeneous graph from both the intra-layer perspective and the inter-layer perspective. In the intra-layer,

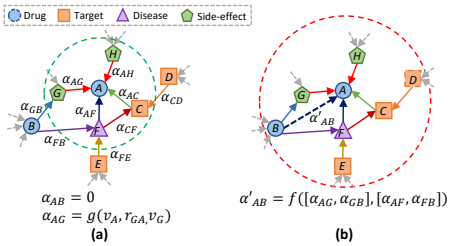

Figure 2: Graph attention diffusion on the biological heterogeneous graph. Let node A be the center node. (a) HGAN first extracts attention scores α between directly connected bioentity node pairs based on representations of end nodes, and edge information. For example, $\alpha_{AB} = 0$ as nodes A and B are not directly connected. The receptive field of node A is one-hop (the green dotted circle). (b) HGAN enhances graph structure learning of node A by considering all paths within its two hops (the red dotted circle) via the diffused attention. For instance, the attention score between nodes A and B is α'_{AB} by combining paths $B \rightarrow G \rightarrow A$ and $B \rightarrow F \rightarrow A$ together.

inspired by the superiority of graph diffusion networks [15, 16, 47] in capturing multi-hop neighboring information, we develop an enhanced graph attention diffusion layer (EGADL) to aggregate neighboring node features within a number of hops for a center node, as illustrated in Figure 2. During this procedure, we first extract attention weights on edges (Figure 2(a)). Subsequently, we further distill attention weights between node pairs that are multiple hops away via an attention diffusion process, which enables information passing from important nodes multiple hops away (represented by the dotted arrow in Figure 2(b)). Mathematically, the attention diffusion places a personalized page rank (PPR) [15, 16, 47] prior on the biological heterogeneous graph edge attention weights. In order to enhance node representation learning and optimally preserve node embeddings while aggregating embeddings from its local neighborhood, we propose a parameter learning module to offer each EGADL a specific teleport probability.

By applying several consecutive EGADLs on the biological heterogeneous graph, we further enlarge receptive fields from the inter-layer perspective. Accordingly, we obtain bioentity node features characterized with rich semantic information. Besides, to improve model expressiveness, we propose a feature combination layer to make the best of bioentity node features mined from different scales. Afterwards, we obtain the representations of drug-target pairs (DTPs) and feed them into the prediction module to identify whether the interactions exist for given DTPs. To the best of our knowledge, we are the first to operate directly on the biological heterogeneous graph for DTI prediction.

The contributions of our paper are summarized as follows.

• We propose a novel heterogeneous graph attention network to leverage long-range dependencies in the biological heterogeneous graph to boost DTI prediction performance. Specifically, we enlarge the receptive fields of HGAN on the biological heterogeneous graph from both the intra-layer and the inter-layer perspectives.

• We propose the enhanced graph attention diffusion layer to explore context-dependent representations for each bioentity node. Particularly, we design a parameter learning module to offer each layer a specific teleport probability, which enables each layer to

optimally preserve node features while aggregating features from its local neighborhood,

• We experiment on two biological heterogenous datasets. HGAN significantly outperforms 15 state-of-the-art methods, achieving the results near to 1 on AUC and AUPR, which verifies the efficacy of HGAN for DTI prediction. Additionally, the experiments show that enlarging receptive fields from the inter-layer perspective (stacking layers) is superior to that from the intra-layer perspective (attention diffusion) for HGAN to achieve promising DTI prediction results.

2 RELATED WORK

2.1 Drug-Target Interaction Prediction

DTI is described as the binding of a drug molecule to a target (usually a protein), in which a drug inhibits or activates a target to treat diseases. Accurate identification of DTIs can considerably accelerate drug discovery process. Most works only employ chemical and genomic data for DTI prediction [14, 28, 33, 34, 55, 58] by modeling drugs and targets in the form of strings, molecular graphs or complexes. However, the relatively limited knowledge about complex relations between the chemical space and the genomic space significantly restricts DTI prediction.

With the rapid development of biological technologies, such as genomics, proteomics, transcriptomics, and metabolomics, huge biological database resources [7, 8, 26, 41, 48] become available, which provides rich information for DTI prediction. Many efforts have been devoted to improve DTI prediction accuracy from heterogeneous biological data using ML-based methods. These methods can be generally categorized into similarity-based, knowledge graph (KG) based, and GNN-based methods. The similarity-based methods [1, 31, 39, 51] hold an assumption that similar drugs are prone to interact with similar targets, and vice versa. They learn drug and target representations from multiple drug- and target-related similarity networks via ML-based models, e.g., random walk and autoencoder (AE). As a result, the learned representations of drugs and targets may not necessarily capture the complex semantic relations between bioentities. Additionally, it is time-consuming to calculate the pairwise similarities for large-scale datasets.

The KG-based techniques associate bioentity node relations with shallow embeddings based on knowledge graph embedding models, e.g., TransE [2], DistMult [20] and ComplEx [44]. For instance, Mohamed et al. [32] and Zhang et al. [54] customized ComplEx [44] to identify DTIs. KGE_NFM [50] integrated representations of drugs and targets learned from DistMult and then fed them into the neural factorization machine (NFM) [18] to realize DTI prediction. Whereas, these methods are deficient in modeling composition relations in the biological heterogeneous graph.

The GNN-based methods [20, 27, 36, 46] extract drug and target representations from multiple drug- and target-related networks using a GCN- or GAT-based model. Some works are the hybrids of GNNs and other strategies. For example, DTI-MGNN [29] integrated the similarity-based method and the GNN-based method to predict DTIs. It viewed DTPs as nodes and constructed a topology graph based on DTIs and a feature graph based on representations learned from drug and target similarity matrices using a sequence of network diffusion and AE. IMCHGAN [27] adopted metapaths [11] and GATs to learn drug and target representations for DTI prediction. However, the design of metapaths is based on meticulous manmade rules and significantly relies on domain knowledge. Also, it is challenging to generalize such an enumerated scheme to large-scale datasets. In general, these GNN-based methods treat bioentities and their relations in an identical manner. They do not highlight different roles performed by bioentities in different relations. Moreover, they are deficient in capturing long-range semantic relations in the biological heterogeneous graph.

HGAN belongs to the GNN-based methods. It is capable of modeling complex bioentity interrelations and exploiting long-range dependencies in biological heterogeneous graph. Meanwhile, HGAN takes into consideration bioentity node features and edge functional information, which enables the learned bioentity representations with rich structural and semantic information.

2.2 Graph Neural Networks

GNNs have become one of the most promising paradigms for learning representations from graph-structured data. They can be generally categorized into spectral-based models, e.g., ChebNet [10], and spatial-based models, e.g., GraphSAGE [17], GCN [23] and GAT [45]. The former utilizes the spectral graph theory to conduct convolution operations on graphs. The latter defines convolutions directly on graphs by aggregating features from neighborhood for each node, which facilitates generalization to unseen nodes or graphs. Particularly, GAT [45] generalizes attention mechanisms into graph data by assigning neighboring nodes with different importance at the feature aggregation step. However, these GNN models only consider direct neighbors for feature aggregation at each layer, and suffer from over-smoothing with deep architectures [4]. Latterly, graph diffusion networks [15, 16] are developed to aggregate multi-hop neighbors in a layer. They combine the strengths of both spatial and spectral methods by approximating the message passing with the PPR [16]. MAGNA [47] further incorporates GAT and graph diffusion, and is able to extend its application to heterogeneous graphs. Nevertheless, the teleport probability used in these graph diffusion networks is empirically set to a fixed value, which restricts their capabilities to learn optimal node features.

We borrow the idea of MAGNA, designing HGAN based on GAT and the graph diffusion mechanism. The core idea is to enhance large-scale structural information learning in the biological heterogeneous graph. Different from MAGNA, HGAN learns teleport probability dynamically in a data-driven manner.

3 METHOD

In this section, we present details of the proposed HGAN. As illustrated in Figure 3, it is constructed with four key components, i.e., node and edge type-specific transformations, graph attention diffusion, feature combination and prediction modules.

3.1 Graph Construction

A heterogeneous graph is defined as a directed graph $G = (V, E, \mathcal{A}, \mathcal{R})$, where $V, E, \mathcal{A}, \mathcal{R}$ represent node set, edge set, node type set and edge type set, respectively, and $|\mathcal{A}| + |\mathcal{R}| > 2$. Each node $v \in V$ is associated with its type mapping function $\phi(v) : V \rightarrow \mathcal{A}$, and each edge $e \in E$ is associated with its type mapping function $\psi(e) : E \rightarrow \mathcal{R}$.

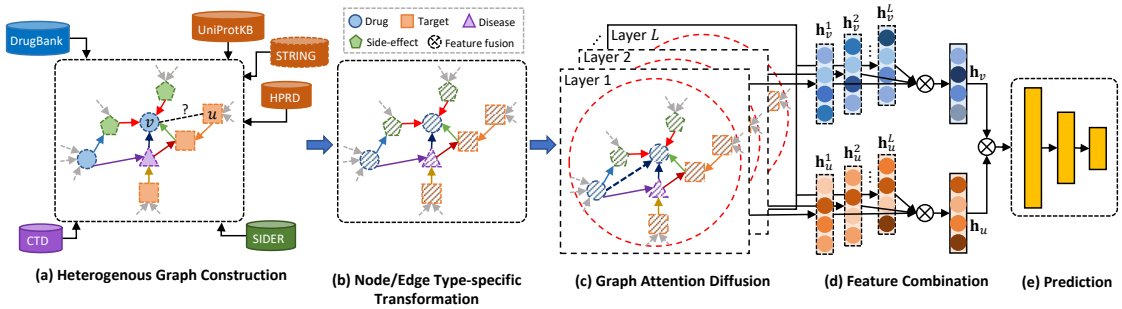

Figure 3: Overall framework of the proposed HGAN. (a) Constructing the biological heterogeneous graph from multiple database resources. The black dotted line denotes the interaction prediction between nodes v and u. (b) Transforming node features and edge type features into the same feature space to eliminate the heterogeneous gap. (c) Aggregating long-range semantic information via graph attention diffusion and layer stacking. (d) Combining multi-scale features mined from different layers for each drug and each target node, and combining drug and target features. (e) Predicting DTIs.

We construct a biological heterogeneous graph from multiple public database resources [7, 8, 26, 41, 48] to model the complex interrelations between diverse bioentities. It is composed of four types of nodes, i.e., drug (D), target (T), disease (I) and side-effect (S), and 10 types of edges/relations, i.e., D-T, D-D, D-I, D-S, T-T, T-D, T-I, I-D, I-T and S-D (see Figure 2(a)). Each bioentity node v_i $(1 \le i \le N_n)$ is represented as a vector $\mathbf{x}_i \in \mathbb{R}^{d_n}$, and each relation type r_j $(1 \le j \le N_r)$ is represented as a vector $\mathbf{r}_j \in \mathbb{R}^{d_r}$, where $N_n = |V|$ and $N_r = |\mathcal{R}|$ denote the number of bioentity nodes and the number of edge types, and d_n and d_r denote the dimensions of each node embedding and each edge type embedding, respectively. Accordingly, we obtain a feature matrix $\mathbf{X} \in \mathbb{R}^{N_n \times d_n}$ for bioentity nodes and a feature matrix $\mathbf{R} \in \mathbb{R}^{N_r \times d_r}$ for edge types, where $\mathbf{x}_i = \mathbf{X}[i,:]$, and $\mathbf{r}_j = \mathbf{R}[j,:]$.

HGAN builds on GAT, while combining the benefits of the diffusion technique. It learns low-dimensional representations of bioentity nodes and relation types to facilitate the DTI prediction task.

3.2 Node/Edge Type-specific Transformation

Due to heterogeneity, the nodes of drugs, targets, diseases and side-effects lie in different feature spaces and have different feature dimensions. So do the complicated interrelations between bioentities. Then, it is a troublesome to process various bioentities as well as relations in a unified framework. Therefore, for each type of nodes, we apply node type-specific transformation $\mathbf{W}_A$ to project them into the same feature space. Similarly, for each type of relations, we apply edge type-specific transformation $\mathbf{W}_R$ to project them into the same feature space.

$$\mathbf{h}'_i = \mathbf{W}_A \cdot \mathbf{x}_i, \mathbf{o}'_j = \mathbf{W}_R \cdot \mathbf{r}_j, \tag{1}$$

where $\mathbf{h}'_i$ and $\mathbf{x}_i$ are the transformed and initial feature vectors of node v_i, $\mathbf{o}'_j$ and $\mathbf{r}_j$ are the transformed and initial feature vectors of edge type r_j, $A \in \{\mathrm{D,T,I,S}\}$ refers to node types, and $R \in \{\mathrm{D\text{-}T, D\text{-}D, D\text{-}I, D\text{-}S, T\text{-}T, T\text{-}D, T\text{-}I, I\text{-}D, I\text{-}T, S\text{-}D}\}$ refers to edge types. Then, we obtain the projected node feature matrix $\mathbf{H}'$ and the projected edge type feature matrix $\mathbf{O}'$, where $\mathbf{H}'[i,:] = \mathbf{h}'_i$, and $\mathbf{O}'[j,:] = \mathbf{o}'_j$. Node and edge type-specific transformations address the heterogeneity that originates from raw node and edge type features, and hence facilitate the following processes.

3.3 Edge Attention Computation

At each layer l, we first apply the layer normalization (LN) [52] operation to node embeddings and edge type embeddings to stabilize model training and boost model convergence.

$$\tilde{\mathbf{H}}^{(l-1)} = \mathrm{LN}(\mathbf{H}^{(l-1)}), \mathbf{O} = \mathrm{LN}(\mathbf{O}), \tag{2}$$

where $\mathbf{H}^{(0)} = \mathbf{H}'$ is the projected node embeddings, and $\mathbf{O} = \mathbf{O}'$ is the projected edge type embeddings for the first training iteration.

Afterwards, for each edge (v_i, r_k, v_j), we combine the features of nodes v_i and v_j and edge type r_k to compute its importance, $\alpha_{i,k,j}$.

$$\alpha_{i,k,j} = \sigma(\mathbf{q}^{(l)} \cdot (\mathbf{W}_h^{(l)} \tilde{\mathbf{h}}_i^{(l-1)} || \mathbf{W}_t^{(l)} \tilde{\mathbf{h}}_j^{(l-1)} || \mathbf{W}_r^{(l)} \mathbf{o}_k)), \tag{3}$$

where σ is the LeakyReLU activation function, $\mathbf{q}^{(l)}$ is the attention weights of the l-th layer, $\mathbf{W}_h^{(l)}$, $\mathbf{W}_t^{(l)}$ and $\mathbf{W}_r^{(l)}$ are learnable parameters of the l-th layer, $||$ denotes the concatenation operation, $\tilde{\mathbf{h}}_i^{(l-1)} = \tilde{\mathbf{H}}^{(l-1)}[i,:]$ $(1 \le i \le N_n)$, and $\mathbf{o}_k = \mathbf{O}[k,:]$ $(1 \le k \le N_r)$.

By applying Equation (3) on each edge of the biological heterogeneous graph G, we obtain an attention score matrix $\mathbf{S}^{(l)}$. Additionally, the quality of annotations varies across different database resources and there exists redundant and missing information in the biological heterogeneous graph, which potentially impacts the expressiveness of the learned bioentity representations. To prevent over-fitting and strengthen model robustness, we first randomly drop a fraction of edges, and then select top-K edges incident to each node v_i. Namely, each row of $\mathbf{S}^{(l)}$ has K none negative infinite values at most. Afterwards, we apply the softmax function over each row of $\mathbf{S}^{(l)}$ to obtain the normalized attention matrix $\tilde{\mathbf{S}}^{(l)}$

$$\begin{aligned} \mathbf{S}_{i,j}^{(l)} &= \begin{cases} \alpha_{i,k,j}, & (v_i, r_k, v_j) \in E \\ -\infty, & \text{else} \end{cases}, \\ \tilde{\mathbf{S}}^{(l)} &= \mathrm{softmax}(\mathbf{S}^{(l)}), \end{aligned} \tag{4}$$

$\tilde{\mathbf{S}}_{i,j}^{(l)}$ denotes the importance of messages from node v_j to node v_i at layer l, and $\sum_{j=0}^{\infty} \tilde{\mathbf{S}}_{i,j} = 1$.

3.4 Enhanced Graph Attention Diffusion Layer

Attention diffusion. The EGADL aims to capture biological semantics in a long-range locality perspective with the attention

mechanism. Based on the calculated scores $\tilde{\mathbf{S}}^{(l)}$, it distills the attention values between bioentity nodes that are not directly connected and may be multi-hops away in the biological heterogeneous graph. We achieve this via the graph diffusion [16, 47] which is based on the powers of the attention matrix $\tilde{\mathbf{S}}^{(l)}$.

$$\check{\mathbf{S}}^{(l)} = \sum_{m=0}^{\infty} \theta_m \tilde{\mathbf{S}}^m, \tag{5}$$

where θ_m is the decay factor, $\sum_{i=0}^{\infty} \theta_m = 1$, $\theta_m \in [0, 1]$. The superscript of $\tilde{\mathbf{S}}^{(l)}$ is omitted for clarity. θ_m follows the geometric distribution, i.e., $\theta_m = \gamma(1-\gamma)^m$, where $\gamma \in (0, 1)$.

Then, we aggregate bioentity node representations based on the graph attention diffusion operation,

$$\mathbf{H}^{(l)} = \Gamma(G, \tilde{\mathbf{H}}^{(l-1)}, \Theta) = \check{\mathbf{S}}^{(l)} \tilde{\mathbf{H}}^{(l-1)}, \tag{6}$$

where Θ is the learnable parameter set for attention computation.

From Equation (5) and Equation (6), we have the following observations. (1) The powers of the attention matrix, $\tilde{\mathbf{S}}^m$, enlarges the receptive field of the attention from bioentity node v_j to node v_i up to m. It effectively creates attention shortcuts between bioentity nodes that are multiple hops away and may not be directly connected (see Figure 1(b)). (2) The diffused attention weight, $\check{\mathbf{S}}_{ij}^{(l)}$, integrates attention coefficients of different relation paths from bioentity node v_j to node v_i, which enables HGAN to capture biological semantics in a long-range locality perspective for node v_i. (3) Furthermore, attention through each path is also weighted differentially, depending on θ_m and the path length m.

To increase the expressiveness of HGAN, we further extend Equation (6) into the multi-head graph attention diffusion by performing Q independent graph attention diffusions and then concatenating their results.

$$\mathbf{H}^{(l)} = \mathop{\Big\|}_{q=1}^{Q} \mathbf{H}_q^{(l)}, \tag{7}$$

where $\mathbf{H}_q^{(l)} = \Gamma(G, \tilde{\mathbf{H}}^{(l-1)}, \Theta_q)$, $\|$ denotes the concatenation operation, Q is the number of attention diffusion heads, Θ_i represents the trainable parameters of the q-th head.

Approximate computation for attention diffusion. The computation of powers of the attention matrix in Equation (5) is expensive on the large biological heterogeneous graph. To handle this issue, we approximate $\check{\mathbf{S}}^{(l)}\tilde{\mathbf{H}}^{(l-1)}$ in Equation (5) to PPR with the dynamic teleport probability (restart) $\gamma^{(l)}$ by defining a sequence $\mathbf{Z}^{(M)}$, namely, $\check{\mathbf{S}}^{(l)}\tilde{\mathbf{H}}^{(l-1)} = \lim_{M\to\infty} \mathbf{Z}^{(M)}$, where M is the number of hops.

$$\begin{aligned} \mathbf{Z}^{(0)} &= \tilde{\mathbf{H}}^{(l-1)}, \\ \mathbf{Z}^{(m+1)} &= (1-\gamma^{(l)})\tilde{\mathbf{S}}^{(l)}\mathbf{Z}^{(m)} + \gamma^{(l)}\mathbf{Z}^{(0)}, \end{aligned} \tag{8}$$

where $\gamma^{(l)}$ is the teleport probability of layer l learned from graph embeddings. We detail it in the subsequent part.

Proof. For better understanding, we use $\mathbf{Z}^M$ and $\tilde{\mathbf{S}}$ to represent $\mathbf{Z}^{(M)}$ and $\tilde{\mathbf{S}}^{(l)}$, respectively. Then, after M attention diffusion hops,

$$\mathbf{Z}^M = ((1-\gamma^{(l)})^M \check{\mathbf{S}}^M + \gamma^{(l)} \sum_{m=0}^{M-1} (1-\gamma^{(l)})^m \tilde{\mathbf{S}}^m)\tilde{\mathbf{H}}^{(l-1)}, \tag{9}$$

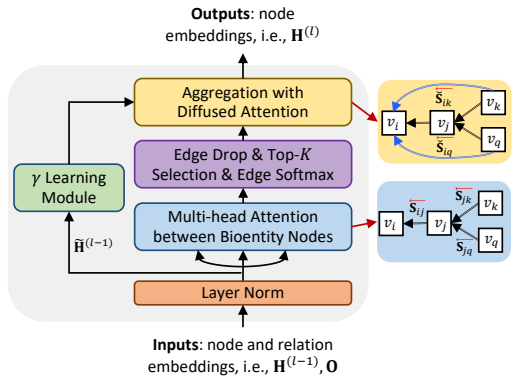

Figure 4: Enhanced graph attention diffusion layer. Each layer consists of the layer normalization, the attention computation, the attention diffusion, and the γ learning modules. As illustrated on the right, context-dependent attentions are achieved via the attention diffusion process, where v_i, v_j, v_k, v_q are nodes in the biological heterogeneous graph.

The term $(1-\gamma^{(l)})^M \check{\mathbf{S}}^M$ converges to 0 when $M \to \infty$ as $\gamma^{(l)} \in (0, 1)$. Thus, $\lim_{M\to\infty} \mathbf{Z}^{(M)} = (\sum_{m=0}^{\infty} \gamma^{(l)}(1-\gamma^{(l)})^m \tilde{\mathbf{S}}^m)\tilde{\mathbf{H}}^{(l-1)} = \check{\mathbf{S}}^{(l)}\tilde{\mathbf{H}}^{(l-1)}$.

Teleport probability γ learning. In existing works [15, 16, 47], the teleport probability $\gamma \in (0, 1)$ is set to a fixed value at all layers, which limits model learning flexibility. Different from them, we design the γ learning module to learn a specific value for each EGADL. The core idea is to optimally preserve the node embeddings while aggregating embeddings from its local neighborhood at each layer. We calculate $\gamma^{(l)}$ by non-linearly transforming the graph embedding (i.e., mean of node embeddings) to a value, and then clamping the value into a fixed range.

$$\begin{aligned} \gamma^{(l)} &= \mathbf{W}_2(\tanh(\mathbf{W}_1 \cdot (\frac{1}{N_n}\sum_{i=1}^{N_n} \tilde{\mathbf{h}}_i^{(l-1)}) + \mathbf{b})), \\ \mathrm{Clamp}(\gamma^{(l)}) &\in [\beta, 1-\beta], \end{aligned} \tag{10}$$

where β is set to 1×10^{-6}. The dimensions of $\mathbf{W}_1$ and $\mathbf{W}_2$ are set to 128 and 1, respectively. Figure 4 presents the architecture of the EGADL of HGAN.

3.5 DTI Prediction

HGAN consists of a stack of L EGADLs. For each bioentity node, its representations extracted from different layers are embedded with diverse semantic information. To take full advantage of the semantic information of a bioentity node captured from different scales, the feature combination layer is proposed to integrate its node representations captured from different layers. Specifically, for drug node v, we obtain its representation from L layers through the following operation,

$$\mathbf{h}_v = \rho(\{\mathbf{h}_v^1, \mathbf{h}_v^2, \cdots, \mathbf{h}_v^L\}), \tag{11}$$

where ρ is the feature combination operation, which can be realized by concatenating, taking the element-wise mean or max of feature vectors of the L layers. Alternatively, we can treat the representation

from the last layer of HGAN, $\mathbf{h}_v^L$, as the representation of drug node v. Similarly, we obtain the representation $\mathbf{h}_u$ for target node u. In section 4.6, we compare DTI prediction performance of HGAN with these feature combination operations.

For each DTP, we achieve its representation by concatenating the drug feature and the target feature, and then feed its representition into the prediction module built with three linear layers with 1024, 512, and 2 neurons to get its prediction result. The binary cross entropy (BCE) loss is utilized to evaluate the differences between the predicted and the ground-truth DTIs. It is formulated as

$$L = -\frac{1}{B}\sum_{b=1}^{B}[y_b \log(y'_b) + (1 - y_b)\log(1 - y'_b)]. \tag{12}$$

where B is the number of training DTPs, $y_b = 1$ if the interaction of the b-th DTP exists, otherwise, $y_b = 0$, and y'_b is the predicted interaction probability of the b-th DTP, which is normalized with the softmax function.

4 EXPERIMENTS

4.1 Datasets

The proposed HGAN is evaluated on two heterogeneous biological datasets. **Hetero-A** is the gold standard dataset introduced in DTINet [31] for DTI prediction with heterogeneous biological data. It is collected from public resources [9, 21, 24, 25], consisting of 12,015 bioentities, i.e., 708 drugs (D), 1,512 targets (T), 5,603 diseases (I) and 4,192 side-effects (S), and six type of connections (1,895,445 interactions in total), i.e., D-T, D-D, D-I, D-S, T-T and T-I. **Hetero-B** is an extension of Hetero-A. It is gathered from October 1 to 15, 2021, from a list of resources [7, 8, 26, 41, 48]. Compared with Hetero-A, Hetero-B (15,322 bioentities and 5,126,875 interactions) contains much more samples for model training and evaluation. The statistical information and resources about hetero-A and Hetero-B are summarized in Table 1. More details about these two datasets are presented in the Appendix A

4.2 Implementation Details

For HGAN, we conduct a 10-fold cross validation procedure for evaluation. The area under the curve (AUC) and the area under the precision-recall curve (AUPR) are employed as the evaluation metrics. We consider all known DTIs as positive samples, and randomly select the same number of unknown DTPs as negative samples. For each fold, 90% positive samples and negative samples are randomly selected as the training set, and the remaining samples are used as the test set for DTI prediction. The Adam is employed as the optimizer with weight decay rate 0.01 and learning rate 0.001. The early stopping with the patience of 30 is utilized. The dimensions of the projected and hidden features are set to 256 and 128, respectively.

4.3 HGAN Model Depth Analysis

The depth of HGAN is determined by the number of attention diffusion hops M and the number of graph attention diffusion layers L. To investigate their impacts on DTI prediction performance, we conduct experiments by varying M from 1 to 5, and L from 1 to 20. The DTI prediction performance of HGAN with different M and L on datasets Hetero-A and Hetero-B are presented in Figure 5.

Table 1: Statistics of datasets Hetero-A and Hetero-B.

Datasets	Types	Items	Numbers	Resources
Hetero-A	Node	D	708	DrugBank (Version 3.0) [24]
		T	1,152	HPRD (Release 9) [21]
		I	5,603	CTD (2013) [9]
		S	4,192	SIDER (Version 2) [25]
	Edge	D-T	1,923	DrugBank (Version 3.0) [24]
		D-D	10,036	DrugBank (Version 3.0) [24]
		D-I	199,214	CTD (2013) [9]
		D-S	80,164	SIDER (Version 2) [25]
		T-T	7,363	HPRD (Release 9) [21]
		T-I	1,596,745	CTD (2013) [9]
Hetero-B	Node	D	2,214	DrugBank (Version 5.1.8) [48]
		T	1,968	UniProtKB (Release 2021_04) [7]
		I	7,205	CTD (2021) [8]
		S	3,935	SIDER (Version 4) [26]
	Edge	D-T	8,750	DrugBank (Version 5.1.8) [48]
		D-D	1,091,870	DrugBank (Version 5.1.8) [48]
		D-I	542,970	CTD (2021) [8]
		D-S	104,629	SIDER (Version 4) [26]
		T-T	456,592	STRING (Version 11.0) [41]
		T-I	2,922,064	CTD (2021) [8]

Note: D, T, S and I denote drug, target, side-effect and disease, respectively.

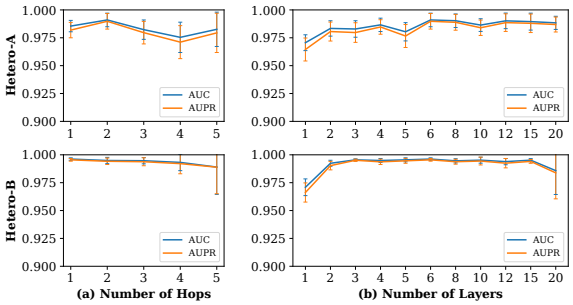

Figure 5: Impacts of (a) different number of attention diffusion hops and (b) different number of graph attention diffusion layers.

Effect of attention diffusion hop numbers M. From Figure 5(a), we observe that HGAN achieves the peak results with a small number of hops, namely, two hops on Hetero-A and one hop on Hetero-B. On Hetero-A, the performance of HGAN first increases at 2 hops. Then, it decreases with the growth of hop numbers. On Hetero-B, the performance has a marginal decline along with the increase of the number of hops M. The possible reason may be that the bioentity node features become smoothing with the growth of M due to the large number of interrelations.

Effect of graph attention diffusion layer numbers L. From Figure 5(b) we find that HGAN achieves significant improvements at 2 layers and 6 layers on Hetero-A. After 6 layers, the performance stabilizes at best results even with 20 layers. On Hetero-B, HGAN has a significant improvement at 2 layers. After that, HGAN present competitive performance with the increase of layers ($2 \leq L < 20$). These results on Hetero-A and Hetero-B suggest that the deep HGAN model is robust and expressive. Comparing the results in Figure 5(b) with these in Figure 5(a), we can conclude that it is more effective for HGAN to enlarge the receptive fields from the inter-layer perspective (stacking multiple layers) rather than from the intra-layer perspective (attention diffusion).

4.4 HGAN Model Robustness Analysis

In this section, we investigate the sensitivity of the proposed HGAN with different parameter settings, including the edge drop rate, and the number of attention heads Q. The DTI prediction results with different parameter settings on datasets Hetero-A and Hetero-B are presented in Figure 6.

Effect of edge drop rate. The quality of annotations varies from diverse biological database resources. It is possible that there exists redundant information in the biological heterogeneous graph. Also, some interactions between bioentities are missing due to limited knowledge and other factors. In order to analyse the robustness of HGAN with these factors, we randomly drop a fraction of edges. From Figure 6(a) we see that the performance of HGAN continually improves and reaches the best result at edge rate 0.2 on Hetero-A. Then, the performance decreases with higher edge drop rate (> 0.2). On Hetero-B, the performance of HGAN has a slight improvement at edge rate 0.2. Then, it remains stable with edge drop rate increases. These results indicates that dropping edges does help for improving model robustness.

Effect of attention head numbers Q. We also explore the performance of HGAN with different number of attention heads. Based on the results in Figure 6(b), we find that the performance of HGAN continually improves with more attention heads on both Hetero-A and Hetero-B. It indicates that multi-head attention mechanism facilitates the improvements of model robustness and expressiveness.

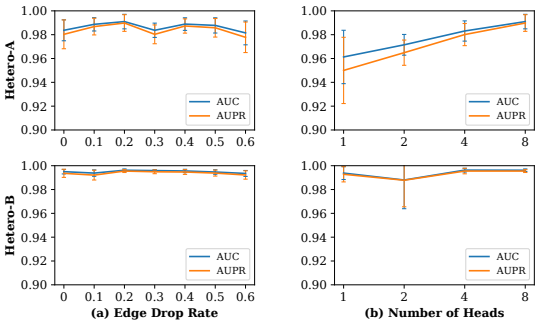

Figure 6: Impacts of (a) different edge drop rates and (b) different number of attention heads.

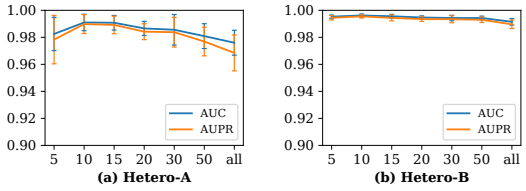

Figure 7: Impacts of different number of neighbors for each bioentity node.

4.5 Graph Width Analysis

The number of different bioentity node interactions from various relations is imbalanced. Also, for a bioentity node, the graph width increases along with the increase of its receptive field. Then, the messages aggregated from minor nodes and relations may interfere or even overwhelm the information useful for bioentity representation learning, leading to suboptimal DTI prediction performance. To investigate the impact of graph width on DTI prediction performance, we choose top-K neighbors for each bioentity node from all of its interactions based on their importance. The prediction results of HGAN with different neighbors for each node are depicted in Figure 7. We find that HGAN reaches its peak performance with 10 neighbors for each node on Hetero-A. Then, its performance declines continually. Particularly, HGAN achieves the lowest results by taking all interactions of a node into consideration. Similarly, on Hetero-B, HGAN performs favourably with 10 neighbors for each node against other situations, and achieves the inferior results by considering all interactions of each node. The results demonstrate that selecting top-K neighbors for each node can enhance model robustness and model expressiveness, and minimize impacts of graph noise. Furthermore, from Figure 5 ~ Figure 7, we find that the standard deviations computed on Hetero-B are generally smaller than those computed on Hetero-A. It shows the augmentation of positive samples enhances model robustness for DTI prediction.

For the subsequent experiments, we set $M = 2$, $L = 6$ for Hetero-A, and $M = 1$, $L = 6$ for Hetero-B. The following parameters are set to be the same for them, $K = 10$, $Q = 8$, and edge drop rate 0.2.

Table 2: Ablation studies on datasets Hetero-A and Hetero-B.

Methods	Hetero-A		Hetero-B	
	$AUC_{\pm std}$	$AUPR_{\pm std}$	$AUC_{\pm std}$	$AUPR_{\pm std}$
Var_{mean}	$0.9852_{\pm 0.006}$	$0.9829_{\pm 0.007}$	$0.9922_{\pm 0.002}$	$0.9901_{\pm 0.003}$
Var_{max}	$0.9818_{\pm 0.009}$	$0.9780_{\pm 0.012}$	$0.9806_{\pm 0.008}$	$0.9745_{\pm 0.012}$
Var_{last}	$0.9850_{\pm 0.009}$	$0.9824_{\pm 0.011}$	$0.9934_{\pm 0.003}$	$0.9919_{\pm 0.004}$
Var_{norel}	$0.9861_{\pm 0.004}$	$0.9843_{\pm 0.004}$	$0.9873_{\pm 0.006}$	$0.9849_{\pm 0.008}$
Var_{nolm}	$0.9691_{\pm 0.017}$	$0.9667_{\pm 0.015}$	$0.9498_{\pm 0.003}$	$0.9365_{\pm 0.008}$
Var_{fix}	$0.9749_{\pm 0.006}$	$0.9702_{\pm 0.007}$	$0.9857_{\pm 0.002}$	$0.9814_{\pm 0.003}$
HGAN	$\mathbf{0.9910_{\pm 0.006}}$	$\mathbf{0.9898_{\pm 0.007}}$	$\mathbf{0.9962_{\pm 0.001}}$	$\mathbf{0.9955_{\pm 0.001}}$

4.6 Ablation Study

To validate the rationality of each component of HGAN, we further conduct experiments on the following variants of HGAN, and the experimental results are reported in Table 2. $\mathbf{Var_{mean}}$ takes the element-wise mean of drug (target) feature vectors extracted from L layers. $\mathbf{Var_{max}}$ takes the element-wise max of drug (target) feature vectors extracted from L layers. $\mathbf{Var_{last}}$ extract drug (target) features from the last layer. $\mathbf{Var_{norel}}$ only considers end node features to calculate edge attention scores while neglecting the relation information. $\mathbf{Var_{nolm}}$ are the models that remove the layer normalization operation. $\mathbf{Var_{fix}}$ denotes the variant with the fixed teleport probability ($\gamma = 0.05$) at each EGADL.

From Table 2 we have the following observations. First, HGAN exceeds all of its alternatives on both Hetero-A and Hetero-B. Second, HGAN outperforms Var_{mean}, Var_{max} and Var_{last}. It indicates that multi-scale drug (target) features contain complementary information, and integrating them with concatenation operation benefits DTI prediction. Third, the relation information is beneficial for DTI prediction improvement. Specifically, HGAN outperforms Var_{norel} by 0.0049/0.0055 on Hetero-A and by 0.0089/0.0106 on Hetero-B

over the metrics of AUC/AUPR. Fourth, the layer normalization play a crucial role in boosting DTI prediction performance. Especially, the improvements are 0.0209/0.0231 (AUC/AUPR) on Hetero-A and 0.0464/0.0590 (AUC/AUPR) on Hetero-B. Finally, the layer-specific γ contributes to DTI prediction improvement.

Table 3: Comparison results of HGAN with 15 methods on datasets Hetero-A and Hetero-B.

Methods	Hetero-A		Hetero-B	
	$AUC_{\pm std}$	$AUPR_{\pm std}$	$AUC_{\pm std}$	$AUPR_{\pm std}$
DTINet	$0.8838_{\pm 0.022}$	$0.9024_{\pm 0.017}$	$0.8239_{\pm 0.026}$	$0.8378_{\pm 0.023}$
deepDTnet	$0.7688_{\pm 0.040}$	$0.7744_{\pm 0.040}$	$0.8129_{\pm 0.097}$	$0.8165_{\pm 0.099}$
MEDTI	$0.7421_{\pm 0.038}$	$0.7428_{\pm 0.029}$	$0.6870_{\pm 0.018}$	$0.7153_{\pm 0.021}$
NEDTP	$0.9115_{\pm 0.021}$	$0.9193_{\pm 0.021}$	$0.9175_{\pm 0.007}$	$0.9221_{\pm 0.006}$
MultiDTI	$0.8951_{\pm 0.016}$	$0.9126_{\pm 0.015}$	$0.8783_{\pm 0.018}$	$0.8883_{\pm 0.015}$
NeoDTI	$0.8690_{\pm 0.052}$	$0.8821_{\pm 0.047}$	$0.9072_{\pm 0.031}$	$0.9078_{\pm 0.032}$
IMCHGAN	$0.8862_{\pm 0.021}$	$0.9000_{\pm 0.021}$	$0.8706_{\pm 0.021}$	$0.8863_{\pm 0.018}$
DTI-MGNN	$0.8969_{\pm 0.026}$	$0.8940_{\pm 0.025}$	$0.8783_{\pm 0.018}$	$0.8883_{\pm 0.015}$
TransE	$0.7435_{\pm 0.035}$	$0.7974_{\pm 0.025}$	$0.6720_{\pm 0.018}$	$0.6777_{\pm 0.019}$
TransE-NFM	$0.9247_{\pm 0.014}$	$0.9286_{\pm 0.014}$	$0.9161_{\pm 0.008}$	$0.9124_{\pm 0.009}$
DistMult	$0.7436_{\pm 0.035}$	$0.7975_{\pm 0.025}$	$0.7441_{\pm 0.012}$	$0.7677_{\pm 0.013}$
KGE_NFM	$0.9250_{\pm 0.014}$	$0.9285_{\pm 0.015}$	$0.9149_{\pm 0.008}$	$0.9151_{\pm 0.009}$
ComplEx	$0.7436_{\pm 0.035}$	$0.7977_{\pm 0.025}$	$0.7440_{\pm 0.012}$	$0.7676_{\pm 0.013}$
ComplEx-NFM	$0.9242_{\pm 0.013}$	$0.9265_{\pm 0.017}$	$0.9161_{\pm 0.009}$	$0.9167_{\pm 0.011}$
MAGNA	$0.9752_{\pm 0.006}$	$0.9696_{\pm 0.008}$	$0.9826_{\pm 0.002}$	$0.9779_{\pm 0.003}$
HGAN (Ours)	$\mathbf{0.9910_{\pm 0.006}}$	$\mathbf{0.9898_{\pm 0.007}}$	$\mathbf{0.9962_{\pm 0.001}}$	$\mathbf{0.9955_{\pm 0.001}}$

4.7 Comparison with Baselines

HGAN is compared against 15 baselines, including 5 similarity-based methods (DTINet [31], deepDTnet [51], MEDTI [39], NEDTP [1], and MultiDTI [57]), 3 GNN-based methods (NeoDTI [46], IMCHGAN [27] and DTI-MGNN [29]), 6 KG-based methods (TransE [2], DistMult [20], ComplEx [44], KGE_NFM [50], TransE-NFM and ComplEx-NFM), and 1 heterogenous graph embedding method (MAGNA [47]). Brief introductions about these baselines are provided in the Appendix B. For fair comparison, all these baselines are re-implemented using their released code by adopting the same training and test set as HGAN, and evaluated with the 10-fold cross validation procedure.

The comparison results are listed in Table 3. It shows that HGAN yields the highest AUC/AUPR performance (0.9910/0.9898 on Hetero-A, 0.9962/0.9955 on Hetero-B) over all baselines, with improvements of 0.0158/0.0202 and 0.0136/0.0176 over the corresponding best results of baselines on Hetero-A and Hetero-B, respectively, which demonstrate its superiority in DTI prediction. The promising results of HGAN can be contributed to the following aspects. Firstly, compared with the similarity-based methods, HGAN is capable of modeling complex interrelations between bioentities and learning node features in an end-to-end manner. However, the similarity-based methods (i.e., DTINet, deepDTnet, MEDTI, NEDTP) are deficient in modeling complex data correlations. Additionally, they separate feature learning from the prediction task, resulting in suboptimal solutions for DTI prediction as the learned representations of drugs and targets may deviate from the prediction task.

Secondly, compared with the GNN-based methods (e.g., NeoDTI and IMCHGAN), HGAN directly performs on the biological heterogeneous graph rather than separating the biological heterogeneous graph into multiple drug- and target-related networks. Also, HGAN extracts edge attentions using both nodes features and edge features, and selects top-K neighbors for each node. Thus, it is able to dynamically concentrate on important nodes and relations according to their importance. Moreover, HGAN broadens the receptive field from both intra-layer perspective and inter-layer perspective, which enables it to capture long-range semantics in the biological heterogeneous graph.

Thirdly, compared with the KG-based methods, HGAN enables the learning of deep bioentity node representations. It also enables graph structural information learning via graph attention diffusion.

Lastly, compared with MAGNA, HGAN assigns a specific teleport probability for each graph attention diffusion layer, which encourages it maintain the tradeoff between node features and features from node neighbors in a data-driven manner. In addition, HGAN makes the most of node features extracted from multiple scales, which enhances its expressiveness.

5 CONCLUSION

In this paper, we have proposed HGAN based on attention mechanism and diffusion technique to capture complex structures and rich semantics in the biological heterogeneous graph for DTI prediction. In order to make the best of diverse information from different biological database resources, we construct the biological heterogeneous graph with both node information and edge type information. Then, we transform node features and edge type features into the same feature space in order to eliminate the heterogeneity gap originated from raw data. Next, we utilize the graph attention diffusion module to learn deep representations for each bioentity node. During this procedure, we learn a specific teleport probability for each layer. Also, we develop the feature fusion layer to combine multi-scale bioentity node representations. We have conducted extensive experiments on two biological heterogeneous datasets, and HGAN outperforms all state-of-the-art methods on metrics AUC and AUPR, which demonstrates its effectiveness in DTI prediction.

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A DATASETS

The proposed HGAN is evaluated on two heterogeneous biological datasets. **Hetero-A** is the gold standard dataset introduced in DTINet [31] for DTI prediction with heterogeneous biological data. It is collected from public resources, consisting of 12,015 bioentities, i.e., 708 drugs (D), 1,512 targets (T), 5,603 diseases (I) and 4,192 side-effects (S), and six connections, i.e., D-T, D-D, D-I, D-S, T-T and T-I. Drugs, targets, diseases and side-effects are gathered from DrugBank (Version 3.0) [24] released in 2010, HPRD (Release 9) [21] released in 2010, CTD (2013) [9] and SIDER (Version 2) [25], respectively. Interactions of D-T and D-D are extracted from DrugBank (Version 3.0), P-P interactions are downloaded from the HPRD (Release 9), associations of D-I and P-I are extracted from the CTD (2013), and D-S associations are gathered from SIDER (Version 2). The processing details of Hetero-A can be referred to [31].

Over the last decade, a large number of novel DTIs as well as other interactions or associations have been discovered, but they

were not fully explored in [31]. The scarcity of positive samples (e.g., DTIs) not only induces errors in data modeling process, but also hides a great risk of false negatives during model evaluation, leading to unknown bias between predictions and the actual results. In light of this, we expand Hetero-A to **Hetero-B** from a list of resources [7, 8, 26, 41, 48], obtaining 15,322 bioentities. Particularly, we collect 2,214 unique drugs from DrugBank (Version 5.1.8) [48]. All the drugs are small molecule drugs and marked "annotated" and "approved". The HPRD database has not been updated since 2010. Hence, we gather targets from UniProtKB (Release 2021_04) [7], wherein, we focus on protein targets in *Homo sapiens (Human)* and filter out targets that are not marked "Reviewed", obtaining 1,968 targets. We assemble 7,205 diseases from CTD (2021) [8] and 3,935 side-effects from SIDER [26], respectively. Note that, we only collect side-effects labeled with PT (preferred term). The interactions of D-T and D-D are obtained from DrugBank (Version 5.1.8), the associations D-I and T-I are compiled from CTD (2021), the D-S associations are gathered from SIDER (Version 4.1) [26], and the P-P interactions are collected from STRING (Version 11.5) [41], wherein P-P interactions include direct (physical) and indirect (functional) interactions.

Hetero-B is collected from October 1 to 15, 2021. It should be noticed that DrugBank, UniProtKB and CTD is updated daily, every eight weeks and monthly, respectively. The STRING (Version 11.5) is released on August 12, 2021. The SIDER is released on October 21, 2015, and its version is linked to the corresponding STITCH version [42] whose latest version is released in 2015. The statistical information and resources about hetero-A and Hetero-B are summarized in Table 1.

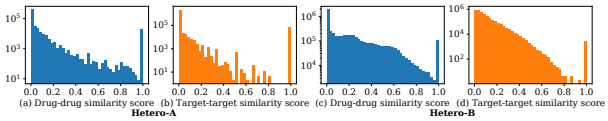

Figure 8: Similarity distributions of DDPs and TTPs on Hetero-A and Hetero-B. The vertical axes are set to the logarithmic scale. The bar at 1.0 of the X-axis in each plot denotes the self-similarity distributions of drugs and targets.

Redundancy check of drugs and targets. High redundant data set could impair model generalization ability. To this end, we check the redundancy of drugs and targets, where the redundancy is measured by their similarity score distributions. Particularly, we calculate the Tanimoto score [43] from drug SMILES (simplified molecular-input line-entry system) strings and use it as the structural similarity for each drug-drug pair (DDP). We compute the normalized Smith-Waterman score [49] from protein sequences and use it as the similarity of each target–target pair (TTP).

Figure 8 shows the similarity distributions of DDPs and TTPs on datasets Hetero-A and Hetero-B. We see that the majority of drug-drug similarity scores are around zeros (about 95.87% in Hetero-A and 88.30% in Hetero-B of DDPs have similarity scores smaller than 0.5). In addition, only about 2.83% in Hetero-A and 0.18% in Hetero-B of TTPs have similarity scores larger than 0.5, and the majority of the scores are close to zeros. Thus, we can conclude that the redundancy of drugs and targets in both Hetero-A and Hetero-B is negligible.

B BASELINES

In this part, we present brief introductions about the approaches that are compared against HGAN.

DTINet [31] extracts drug and target representations from drug- and target-related similarity networks through random walk with restart (RWR) with a dimension reduction scheme. Then, the inductive matrix completion (IMC) is adopted to predict novel DTIs.

deepDTnet [51] converts drug- and target-related similarity networks into drug and target representation matrices via RWR and positive pointwise mutual information (PPMI) techniques. Afterwards, AE models are applied to each matrix to learn drug (target) features separately.

MEDTI [39] is similar to deepDTnet [51]. The difference is that in MEDTI, multiple drug (target) representation matrices are combined together to learn drug (target) features using AE.

NEDTP [1] extracts drug and target features through the Word2Vec algorithm. Then, it constructs the gradient boosting decision tree (GBDT) to predict new DTIs based on learned features.

MultiDTI [57] exploits drug and target representations from sequences using convolutional layers with residual connections, respectively. After that, it maps the representations of drugs, targets, side effects and disease nodes in the heterogeneous network into a common space.

NeoDTI [46] integrates drug (target) representations learned from different relation networks using GCNs. Then, IMC is adopted to predict novel DTIs.

IMCHGAN [27] adopts a two-level neural attention mechanism approach to learn drug and target feature representations from the DTI heterogeneous network respectively. Subsequently, the learned features are fed into IMC to obtain the prediction results of DTIs.

DTI-MGNN [29] first extracts low-dimensional representations of drugs and targets using RWR followed by denoising AE from heterogeneous networks. Then, they regarded DTPs as nodes and constructed topology graph based on known DTIs and feature graph based on learned drug and target features. Afterwards, two GATs and GCNs were employed to explore topological structure and semantic information in DTP representations.

KGE_NFM [50] first utilizes DistMult [20] to extract low-dimensional representations for bioentities. Then, it integrates the learned representations with drug and target structural information through the neural factorization machine (NFM) [18]. Afterwards, the feed-forward neural network is employed to yield DTI prediction. We obtain **ComplEx-NFM** by replacing DistMult in KGE_NFM with ComplEx since Mohamed et al. [32] and Zhang et al. [54] do not release their code. Similarly, we obtain **TransE-NFM**. Accordingly, TransE, ComplEx and DistMult are models that remove NFM.

MAGNA [47] incorporates the attention and the graph diffusion techniques to learn multi-hop context information for each node. Each layer consists of attention computation, attention diffusion, layer normalization, feed-forward layers, and two residual connections. We tailor it for DTI prediction by extracting drug representations and target representations from its last graph attention diffusion layer, concatenating the extracted representations, and then feeding the concatenated result into the prediction module that is the same as HGAN. The teleport probability of MAGNA is fixed to 0.05.

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